

ImpairAssist

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Chapter 1

Introduction

This document serves as a project proposal for ImpairAssist which is an application for audibly impaired people so that they can communicate with the general public by utilizing the real-time captions that our app generates.

1.1 Problem Domain

Throughout the project, our main focus will be speech recognition and enhancing the quality of speech-to-text models. Speech recognition is the ability of computers to identify words or sentences and convert them into a textual format. Speech recognition is a vast domain and a lot of work has already been done on it with continuing work and advancements in the field, till date. Majority of the approaches to speech recognition only focuses on one aspect that is the audio i.e the speech itself, while completely ignoring the visual aspect of speech such as facial cues and lip movements. Leading to erroneous text representations when a different factors such as background noise are involved. For this project we will be addressing this ignoring of the visual input and transitioning towards an approach that incorporates the visual input with the audio input to produce robust text representations of a given speech. This technique of combining audio and visual features is known as Audio-Visual-Speech-Recognition or AVSR.

By integrating both audio (spoken words) and visual (lip movements) inputs, AVSR technology seeks to increase speech recognition accuracy and make speech-to-text more robust. AVSR improves speech recognition by leveraging visual cues, such as lip, mouth, and facial movements, which can offer crucial added contextual cues about speech content, compared to traditional speech recognition systems that only use audio. Therefore AVSR is a better and overall more robust approach to speech recognition as it takes into account a greater number of features of speech.

1.2 Research Problem Statement

People with hearing impairments face a daily struggle to communicate with the general public, they face substantial hurdles on a day to day basis because of the communication gap and the inability to hear, as well as not being able to understand what the other person might be saying. While many hard-of-hearing people rely on lipreading and the movement of lips, this alone does not suffice or encompass all possible scenarios either, such as, rapid speech and complex language. Still the current speech recognition systems, for the majority part, rely on the audio inputs, discarding the visual inputs like, lip movements, facial features and other nonverbal cues that make a sentence whole and contextfull. These models and systems do not perform up to the mark when factors like background noise, poor audio quality and audio-corruption are involved. Due to this, hearing impaired people can't not fully rely on speech to text models to communicate as the text generated by them is not very robust or reliable.

Our focus is to combine the two main aspects that are, audio input and visual input. Audio input includes the speech itself and visual input includes the lip movement of the speaker. The approach we will be taking to achieve this is using separate models for audio and visual inputs and then combining the two models using a form of Model Fusion. This would consist of combining the Audio-Speech-Recognition model's output with the Visual-Speech-Recognition model's output to produce a more robust text representation of speech.

Our software and research aims to find an approach to generate robust, reliable and accurate captions by utilizing both audio and visual inputs so that the audibly impaired can easily communicate with anyone without fear of miscommunication. Our software will address the shortcomings of other speech recognition and caption generation software by catering to the problems that cause other software to give erroneous results. ImpairAssist will cater to problems such as background noise affecting speech recognition, audio corruption, video blurring and other problems. After addressing the above issues ImpairAssist will generate captions in real time.

Chapter 2

Literature Review

2.1 Related Research

2.1.1 LipFormer: Learning to Lipread Unseen Speakers based on Visual-Landmark Transformers (2023)

2.1.1.1 Author:

Feng Xue, Yu Li, Deyin Liu, Yincen Xie, Lin Wu, Richang Hong

2.1.1.2 Summary:

This paper introduces a state of the art multi-modal approach to Visual Speech Recognition called LipFormer. The approach focuses on using the changes of visual landmarks to improve lip reading performance on unseen speakers. Lipreading models perform really well when interpreting overlap speakers, but on unseen speakers, the models fail to produce accurate results due to the limited number of speakers in datasets. LipFormer seeks to improve generalization across a range of speakers by utilizing a two-stream architecture that combines visual features from both lip motion and facial landmarks. By efficiently aligning and fusing these modalities, the framework's distinct self-attention and cross-attention mechanisms enable it to outperform conventional lipreading techniques on benchmark datasets. So this paper proposes a solution to above described problem by making use of visual landmarks

2.1.1.3 Critical Evaluation:

The approach of using multi-modal inputs and the visual changes in facial landmarks is innovative. However, it fails to address its targeted goal of complete Speech Recognition. The model also doesn't evaluate on real-world noisy environments extensively enough. The aspect of Audio-Speech Recognition is also something that is ignored. Although LipFormer presents a promising step forward in lipreading technology, these issues should be resolved, and looking into the possibility of integrating it with audio elements would strengthen its overall usefulness and robustness in practical applications.

2.1.1.4 Relationship to Proposed Research Work:

This paper is highly relevant to our research due to the fact that the model also uses a multi-modal approach and also for the fact that it uses LipNet to perform the Visual Speech Recognition. Moreover, the insights of the paper on the transformer architecture for unseen speaker generalization can help prove very useful when designing our model.

2.1.2 End-To-End Audio-Visual Speech Recognition With Conformers (2021)

2.1.2.1 Authors:

Pingchuan Ma, Stavros Petridis, Maja Pantic

2.1.2.2 Summary of the research item

The authors of this paper discuss and propose a way to maximize the performance of traditional Audio-Visual speech recognition models by introducing an end-to-end model based on conformer architectures. The model is a hybrid CTC/Attention model based on ResNet-18 and Conformer, that can be trained in an end-to-end way, which is the main characteristic of the model since pre-computed visual features and audio do not yield the same result as this new, up and coming approach does.

Additionally the paper has also done a review of the model, tested on the most widely available datasets i.e LRS2 and LRS3, for speech-level recognition. The model raises the state-of-the-art performance significantly for different combinations of experiments, audio-only, visual-only and audio-visual experiments on both datasets, even surpassing the results of methods trained on much larger datasets.

2.1.2.3 Critical analysis of the research item

This end-to-end model based on conformer architecture caters to both audio and visual aspects of speech recognition enhancing robustness of the text prediction. The model's architecture is such that it does not falter even in noisy conditions. The architecture, an end-to-end model, takes the audio waveform and raw pixels and uses them to extract features directly. A few changes were made that greatly boost performance of the audio-visual model, to be specific, they train the model by integrating the feature extraction with the hybrid CTC/attention back-end. All these changes have increased the state-of-the-art performance by a mile.

While this conformer based AVSR model is one that could revolutionise this field, it can be computationally exhaustive, which might limit the practicality of usage in real-time applications.

2.1.2.4 Relationship to the proposed research work

We aim to work on multi-model approaches and Audio-Visual speech recognition models for our project and this paper discusses an approach to improve the result and robustness of traditional AVSR models, it also discusses model fusion, both of which are highly relevant to our aim. The datasets chosen in the paper are also the same as the ones we plan to work with during our research.

2.1.3 BERT: A Review of Applications in Natural Language Processing and Understanding (2021)

2.1.3.1 Author:

M. V. Koroteev

2.1.3.2 Summary of the research item

The paper provides an in-depth analysis of BERT(Bidirectional Encoder Representations from Transformers), a transformer based model that has completely changed the landscape of various natural language processing(NLP) tasks. The paper goes in depth on how BERT's bi-directional approach to context analysis helps it to understand the data better hence improving its performance. It outperforms previous benchmark models in tasks such as: text classification, entity recognition, and question answering. It also highlights the training and fine-tuning processes for the BERT too.

2.1.3.3 Critical analysis of the research item

BERT has had great success in the field of NLP, being praised as an evolutionary technology to revolutionise the field. The main key to BERT's success is the use of Bi-Directional Encoders rather than a Uni-Directional encoder. This technique helps it to capture the context of the sentences when, for example, asked questions. It also helps predict the likelihood of words as it has awareness of both sides of the context. Another reason for BERT's popularity is due to the fact that it can be used for many diverse uses. This allows it to be trained or fine-tuned for almost any NLP application and applications from other fields too.

However, like any technology, BERT has its drawbacks and the authors have also highlighted it in their research too. The paper spends detailed amount of time going over the capabilities of BERT but fails to mention its performance in regards to practical deployments. It overlooks sections such as latency, scalability, and energy consumption. Focusing only on its capabilities in controlled benchmarks.

2.1.3.4 Relationship to the proposed research work

This research item is relevant to our work because to use BERT to perform multi-modal fusion, we need to have a better understanding of how it is trained, how it functions and what to expect as results. This research paper helps us to gain an in-depth understanding of BERT and its functionalities and limitations.

2.1.4 Cross-Modality Gated Attention Fusion for Multimodal Sentiment Analysis (2022)

2.1.4.1 Authors:

Ming Jiang, Shaoxiong Ji

2.1.4.2 Summary of the research item

This paper goes in depth of using multi-modal inputs to enhance sentiment analysis. The paper proposes the use of text, audio and visual signals to perform sentiment analysis on the speaker. However, rather than training multiple models and combining their results using post-processing. The paper suggest using multi-modal fusion or to be precise a Cross-Modality Gated Attention Fusion (CM-GAF) mechanism. The CM-GAF can focus on different important features in any of the modalities at any given time.

The model uses LSTM layers to capture the sequential features of the data and then uses

gated attention mechanism to learn to prioritize the most useful features for each modality while ignoring or reducing the importance for others..

2.1.4.3 Critical analysis of the research item

The model handles the use of multi-modal inputs quite well, it combines and helps the input streams to interact with each other allowing it to share valuable features along to each other. Furthermore the cross-modality gated attention mechanism allows the model to intelligently focus on relevant features just like the human brain.

The model while innovative in its approach does not come without its drawbacks. The use of cross-modality attention and gated mechanisms may increase the computational complexity significantly. Another important factor that was overlooked by the authors is the understanding of how each modality affects the output or even if it does. This would require extensive testing to understand which modality has what kind of on the output and the size of its effect too.

2.1.4.4 Relationship to the proposed research work

In our research Cross-modal Attention with Transformer-based Fusion is one of the techniques that comes to close to the results that we want to achieve. It not only successfully fuses the modalities but also allows for intelligent thinking of how they should be merged.

2.1.5 SyncVSR: Data-Efficient Visual Speech Recognition with End-to-End Crossmodal Audio Token Synchronization (2024)

2.1.5.1 Authors:

Young Jin Ahn, Jungwoo Park, Sangha Park, Jonghyun Choi, Kee-Eung Kim.

2.1.5.2 Summary:

SyncVSR aims to improve the quality of videos by leveraging the temporal coherence between frames. It synchronizes multiple frames in a video sequence to enhance their resolution, providing a higher quality output than traditional single-frame super-resolution methods.

2.1.5.3 Critical Evaluation:

The cross modal synchronisation between the audio and visual inputs makes the approach much better in terms of the quantity of data. However the reliance on synchronisation limits the scope of the approach in situations where inputs from either of the streams are absent or distorted heavily. Furthermore while the approach is great in situation where there is lack of data, it is not well adapted for unseen speakers.

2.1.5.4 Relationship to Proposed Research Work:

SyncVSR primarily synchronizes the cross-modal audio tokens to allow the model to process data in sync. Our use of BERT also hopes to achieve the same functionality to allow for smoother and data efficient processing. The fusion process and synchronization between modalities in your model can be improved by implementing the cross-modal synchronization and data efficiency techniques described in this work.

2.1.6 Adapted Multimodal BERT with Layer-Wise Fusion for Sentiment Analysis (2023)

2.1.6.1 Authors:

Odysseas S. Chlapanis, Georgios Paraskevopoulos, Alexandros Potamianos.

2.1.6.2 Summary:

This paper leverages BERT language model for an advanced approach to a multi-modal sentiment analysis. It uses a multi-modality consisting of audio, visuals and text to get a more contextually accurate and more robust sentiment prediction. It uses the approach of keeping the weights in the pre-trained BERT model unchanged by freezing these layers and adding additional adapter and fusion layers to incorporate the different modalities. This paper utilizes a layer-wise fusion approach which ensures that fusion of modalities occurs at multiple layers of BERT, which allows the model to process each of the modalities differently and enhances its ability to capture complex interactions between text, visual, and acoustic data.

2.1.6.3 Critical Evaluation:

The Adapted Multimodal BERT (AMB) achieves parameter efficiency by freezing BERT's parameters and training lightweight adapter modules, reducing computational demands

while maintaining strong performance. Its adapter-based fusion strategy effectively combines text, audio, and visual data at intermediate layers, leveraging BERT’s language knowledge for improved multimodal sentiment analysis. The model’s success on the CMU-MOSEI dataset shows enhanced accuracy and robustness to noisy data, making it practical for real-world use.

However, the model’s reliance on the CMU-MOSEI dataset raises concerns about generalization to other domains or tasks, leaving its broader applicability unexplored. The fusion strategy is effective but it remains relatively simplistic compared to more advanced cross-model techniques that could handle modality-specific challenges, such as data shifts or temporal dynamics, better. Moreover, the model’s performance is heavily dependent on text as the dominant modality, which may limit the value it derives from audio and visual inputs, especially when textual data is insufficient or inaccurate.

2.1.6.4 Relationship to Proposed Research Work:

The paper under discussion is highly relevant to our research as it explores multi-model fusion, similar to our approach of combining audio and visual inputs for speech recognition. By leveraging an adapter-based fusion strategy, AMB improves performance under noisy conditions, which aligns with our goal of enhancing robustness in speech-to-text models for hearing-impaired users. We can apply insights from AMB’s method of integrating different data types to better fuse audio and visual cues, improving accuracy in challenging environments like those with background noise or audio corruption.

2.1.7 Multimodal Multi-loss Fusion Network for Sentiment Analysis (2024)

2.1.7.1 Authors:

Zehui Wu, Ziwei Gong, Jaywon Koo, Julia Hirschberg

2.1.7.2 Summary of the research item

The paper presents a model that evaluates multiple data modalities (text, audio, visual). This model uses specific encoders for each modality to extract all key features then integrates these features to a single representation. Their key innovation is the use of multi-loss functions used in early or late fusions.

This approach focuses on improving each modalities independently using their respective loss functions, hence improving the model’s learning process. Then moving onto fusing of the input streams using a combination of early (for feature extraction) and late (using

pooling and averaging) fusion.

2.1.7.3 Critical analysis of the research item

The paper provides new and interesting way of integrating multiple data types. The process does not require any major changes when required to change the data types too. The use of multi-loss functions is a very viable addition, as it allows each modalities features to be learned by the model separately. Allowing the model to boost it's performance while minimizing their respective weaknesses.

While the multi-loss functions approach is powerful, it increases the complexity of the training process quite enormously. If there are situations with limited computational resources, the training process is simple infeasible. Along with that, the architecture of the model is quite complex and the combinations of multiple loss-functions and the combinations modalities make it quite tough to understand the effect of what each modality contributes to the final result.

2.1.7.4 Relationship to the proposed research work

The research paper's exploration of early and late fusion techniques, along with its multi-loss optimization strategy, is quite close to the process that we have proposed. It also helps us understand the complications of using techniques like early and late fusion when fusing modalities.

2.1.8 Robust Self-Supervised Audio-Visual Speech Recognition. (2022)

2.1.8.1 Authors:

Bowen Shi, Wei-Ning Hsu, Abdelrahman Mohamed.

2.1.8.2 Summary:

The paper introduces a new and robust approach to AVSR via a self supervised learning framework. A major benefit of using a self-supervised approach is that it does not require large labelled datasets. As the approach uses both the audio and visual inputs, it can handle aspects such as noisy environments and missing data quite well.

2.1.8.3 Critical Evaluation:

The use of self-supervised is a notable achievement, as it reduces the reliance of large labelled datasets which is a major issue in the field of AVSR. However, this approach does not extensively test its results across unseen speakers, dialects or different languages.

2.1.8.4 Relationship to Proposed Research Work:

This paper is really interesting because it looks at self-supervised methods for audio-visual speech recognition. This is super useful when you don't have a lot of labeled data to work with. Since you're using BERT to combine visual and audio inputs, the self-supervised techniques in this paper could help you learn strong representations from both types of data without needing a ton of supervision. This might boost performance, especially in situations where data is limited.

2.1.9 A Denoising Autoencoder Based on U-Net and Bidirectional Long Short-Term Memory for Multi-Level Random Telegraph Signal Analysis (2024)

2.1.9.1 Authors:

Bowen Deng, HeeBong Yang, Na Young Kim.

2.1.9.2 Summary:

This paper presents the use of a denoising autoencoder model that combines a U-Net and bidirectional LSTM network to perform analysis of multi-level random telegraph signals (RTS) in audio data. The purpose of this model is to denoise the RTS signals, which are commonly found when dealing with nanoscale devices.

2.1.9.3 Critical Evaluation:

The combination of U-Net and BiLSTM offers a powerful architecture for signal denoising, that handles both spatial and temporal features of RTS. However, the model's complexity may result in increased computational cost, potentially limiting its use in real-time scenarios. Additionally, the paper does not explore the model's performance under different noise levels or signal variations.

2.1.9.4 Relationship to Proposed Research Work:

This paper mainly discusses signal analysis with U-Net and BiLSTM architectures, but its denoising autoencoder framework could really help improve the robustness of your audio and visual inputs. If your model encounters noise—like unclear lip movements or noisy audio—the denoising techniques from this paper could guide your pre-processing or post-processing steps. This way, you can enhance the quality of the input data before combining them, which should lead to better overall performance.

2.1.10 A Review of Deep Learning Techniques for Speech Processing. (2023)

2.1.10.1 Authors:

Ambuj Mehrish, Navonil Majumder, Rishabh Bharadwaj, Rada Mihalcea, Soujanya Poria.

2.1.10.2 Summary:

This paper provides an in depth analysis of the recent advances in the DL techniques for different Speech Processing Methods (speech recognition, identifying speakers and speech enhancement) The paper classifies the advancements with different deep learning models, such as CNN, RNN and transformer based architectures. It discusses the advantages, drawbacks and real life application possibilities.

2.1.10.3 Critical Evaluation:

Research on speech recognition has been going on for many years and many advancements have been made but there is still room for improvement as many flaws still exist and the results are still not as accurate as one would want. Further improvements to speech recognition can be made by utilizing the key features, methodologies and architectures of the speech recognition approaches mentioned in the paper to produce better results.

2.1.10.4 Relationship to Proposed Research Work:

This review provides valuable insights for AVSR research by highlighting how deep learning techniques for speech processing can be integrated with visual data. For instance, methods like Wave2Vec, known for their robust feature extraction, could complement visual models such as LipFormer for better multi-modal fusion. The focus on end-to-end

architectures can inspire future AVSR systems to combine audio and visual processing seamlessly, enhancing recognition accuracy. Additionally, the challenges mentioned, like data scarcity and computational demands, are relevant to AVSR, encouraging innovative approaches in areas like self-supervised learning to address these limitations.

2.2 Analysis Summary of research items

Name	Reference	Year	Approach	Key Features	Deficiencies
LipFormer	LipFormer Reference [10]	2023	3DCNN, GRU, LSTM	Multimodal, Visual Landmark Transformer	Undertrained, when the person changes accuracy drops
SyncVSR	SyncVSR Reference [1]	2024	3DCNN, RezNet18, Transformer	Synchronization of audio and visual inputs	Undertrained, angle of face, inefficient with noise
Robust Self-Supervised Audio-Visual Speech Recognition	Robust Self-Supervised Reference [8]	2022	AV-HuBERT	Self-supervised learning via clusters	Does not factor in dialects and accents.
A denoising autoencoder for multi-level random telegraph signal analysis	Denoising Autoencoder Reference [3]	2024	Bidirectional LSTM, DDAE with UNET	Suppresses local and global background noise	Greater inference time.
A Review of Deep Learning Techniques for Speech Processing	Review of Deep Learning Techniques Reference [7]	2023	Wave2Vec, Whisper, HuBERT, Transformer	Comparative analysis and history of majority speech recognition models	Lists the deficiencies of all types of models

Continued on next page

Name	Reference	Year	Approach	Key Features	Deficiencies
BERT: A Review of Applications in NLP and Understanding	BERT: reference [5]	2021	Review of BERT's role in NLP.	Pre-trained transformer, adaptable to various tasks.	High resource demand, lacks interpretability.
Cross-Modality Gated Attention Fusion for Multimodal Sentiment Analysis	Cross-Modality Gated Attention Fusion reference [4]	2022	Gated attention for multimodal fusion.	Adaptive attention for improved sentiment analysis.	Limited to sentiment tasks, less generalizable.
Multimodal Multi-loss Fusion Network	Multimodal Multi-loss Fusion Network reference [9]	2023	Multi-loss fusion for sentiment analysis.	Balanced input fusion, improved accuracy.	High computational cost, difficult optimization.
Adapted Multimodal BERT with Layer-wise Fusion	Adapted Multimodal [2]	2022	Layer-wise fusion of text, audio, and visual modalities.	Gradual fusion, improved multimodal sentiment prediction.	Complex architecture, higher computation overhead.
End-to-end Audio-visual Speech Recognition with Conformers	End-to-end Audio-visual [6]	2021	Conformers, CTC/Attention model.	Efficient audio-visual modeling, improved recognition accuracy.	Limited to audio-visual tasks, high training complexity.

Table 2.1: Summary of various techniques and their references.

Chapter 3

Proposed Approach

3.1 Proposed Solution/Method

3.1.1 Overall Approach:

Our proposed solution consists of multi model approach that will utilize both audio and visual aspects of speech. Our approach will receive input in the form a video and then extract the visual and audio aspects of speech. The visual aspect that is the movement of the lips will be fed to the visual speech recognition module, the audio aspect will be fed to the audio speech recognition module. Then the results of the audio and visual modules will be passed to across attention module that will combine the audio and visual speech recognition results and generate text caption that are more robust than other speech recognition approaches that utilize a single type of input for recognizing speech. The cross attention module will use either model fusion, a bidirectional transformer or a multi-model architecture to combine the models and their results.

3.1.2 Input and Segmentation:

The pipeline will receive input in the form of a video either in real-time or recorded along. The first step would be isolating the visual component from the audio component, basically separating the audio signals in the video and the actual frames in the video. A simple script will be used to extract the audio and and frames from a videos and save them separately so that then can be utilized in the next step of the pipeline.

3.1.3 Preprocessing:

After obtaining the audio and video frames separately we need to perform preprocessing on the input so that the data contains less noise and it is easier to process it and perform speech recognition using this input. The frames will be denoised and sharpened using tools such as openCV and similarly the audio input will be processed to remove noise present in the audio using tools such as librosa.

3.1.4 Deep Learning:

After the two types of inputs have been preprocessed cleaned and noise has been removed from them, they will pass to the next step in the pipeline which is the deep learning model. The audio input will be fed to a audio-speech-recognition model and the video will be fed to a visual-speech-recognition model, both models will use some form of self-attention and cross-attention and other deep learning techniques to produce caption or text representation of the speaker's speech in the video. These outputs from the two models will be fed to module which is the component responsible for combining the results of audio-speech-recognition and visual-speech-recognition.

3.1.5 Model Combining:

After the visual-speech-recognition model and audio-speech-recognition models have given their respective outputs we need to combine them and get an average result of both the models. This is where we need to combine the two models or carry out model fusion, this will be done by either using a model averaging technique or using cross-model fusion techniques or approaches such as BERT. After the two outputs have been combined, we can move to the last step or section of the pipeline which is post-processing and final output.

3.1.6 Post-processing and Output:

After the model has given its output the final step would be post-processing it to remove small grammatical errors in-order to increase the intelligibility of the text out and after post-processing the generated text representation of the speech will be added as captions in the original video that was given to the pipeline as an input.

3.1.7 Evaluation Metrics:

3.1.7.1 Word Error Rate (WER):

Is a measure of the accuracy of a speech-recognition system at the word level that works by calculating the percentage of words in the reference text that are either substituted, inserted, or deleted in the recognized output. It has the following equation.

$$\text{WER} = \frac{\text{Substitutions} + \text{Insertions} + \text{Deletions}}{\text{Total Number of Words in the Reference}}$$

3.1.7.2 Character Error Rate (CER):

It is a measure of the accuracy of a speech-recognition system at the character level which works by calculating the percentage of characters in the reference text that are either substituted, inserted, or deleted in the recognized output. Similar to WER but at character level.

$$\text{CER} = \frac{\text{Substitutions} + \text{Insertions} + \text{Deletions}}{\text{Total Number of Characters in the Reference}}$$

3.1.7.3 Latency:

The measure of the time the model takes to generate the text representation for each sentence and how much delay there is between receiving input and producing an output.

3.1.8 Proposed Solution Flow Diagram

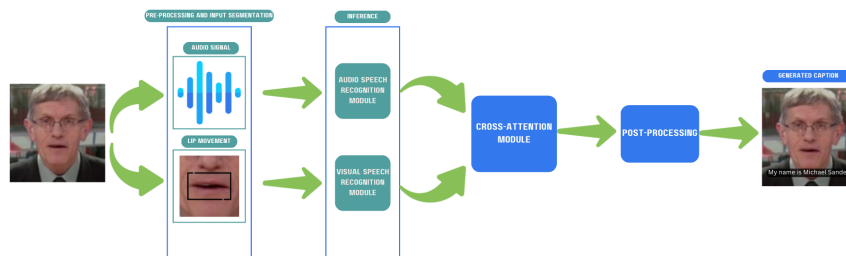


Figure 3.1: Proposed solution diagram

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